

Introductory
Night Sky
photography

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Wide angle imaging of the night sky



The DSLR

Sensitive at low light levels



Method: Camera settings

- f/number
- Exposure time
- ISO setting

Method: Camera settings

- f/number
 - Exposure time
 - ISO setting
- Increase to collect more light...
- ...But objects move across the sky & light pollution.

Exposure Time

Short: $\frac{1}{8}$ seconds



Exposure Time

Longer: 2s

Change the
exposure time
for sunshine or
earthshine



Exposure time: 10 hours!



Light pollution

Typical view of the night sky
from a light polluted location

Canon 10D, 120s at f2.8 & ISO400.

Light pollution

Typical view of the night sky
from a dark location

Canon 10D, 120s at f2.8 & ISO400.

Light pollution from the inefficient lights of Brighton

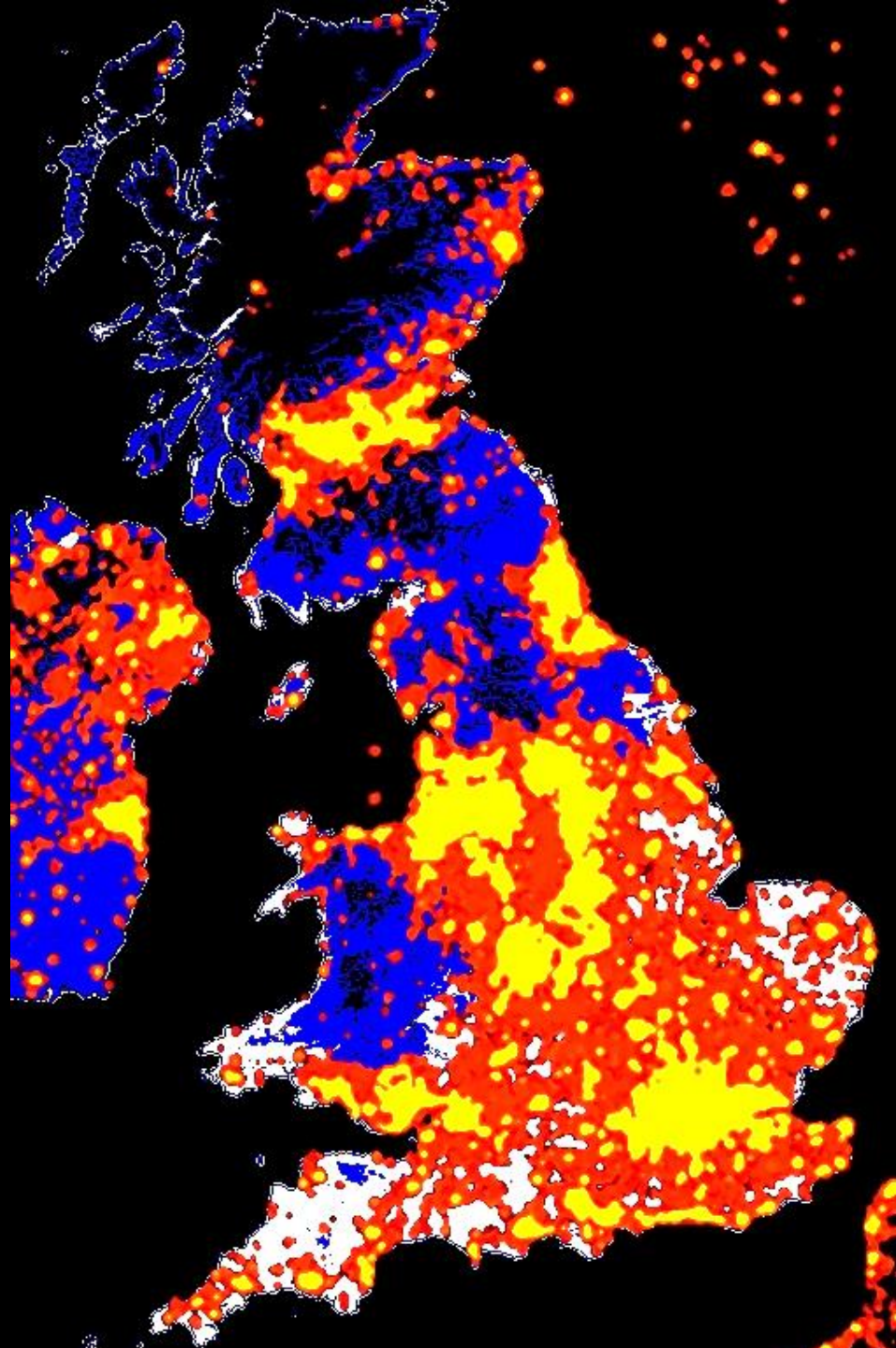


Light pollution from the inefficient lights of Brighton Pier

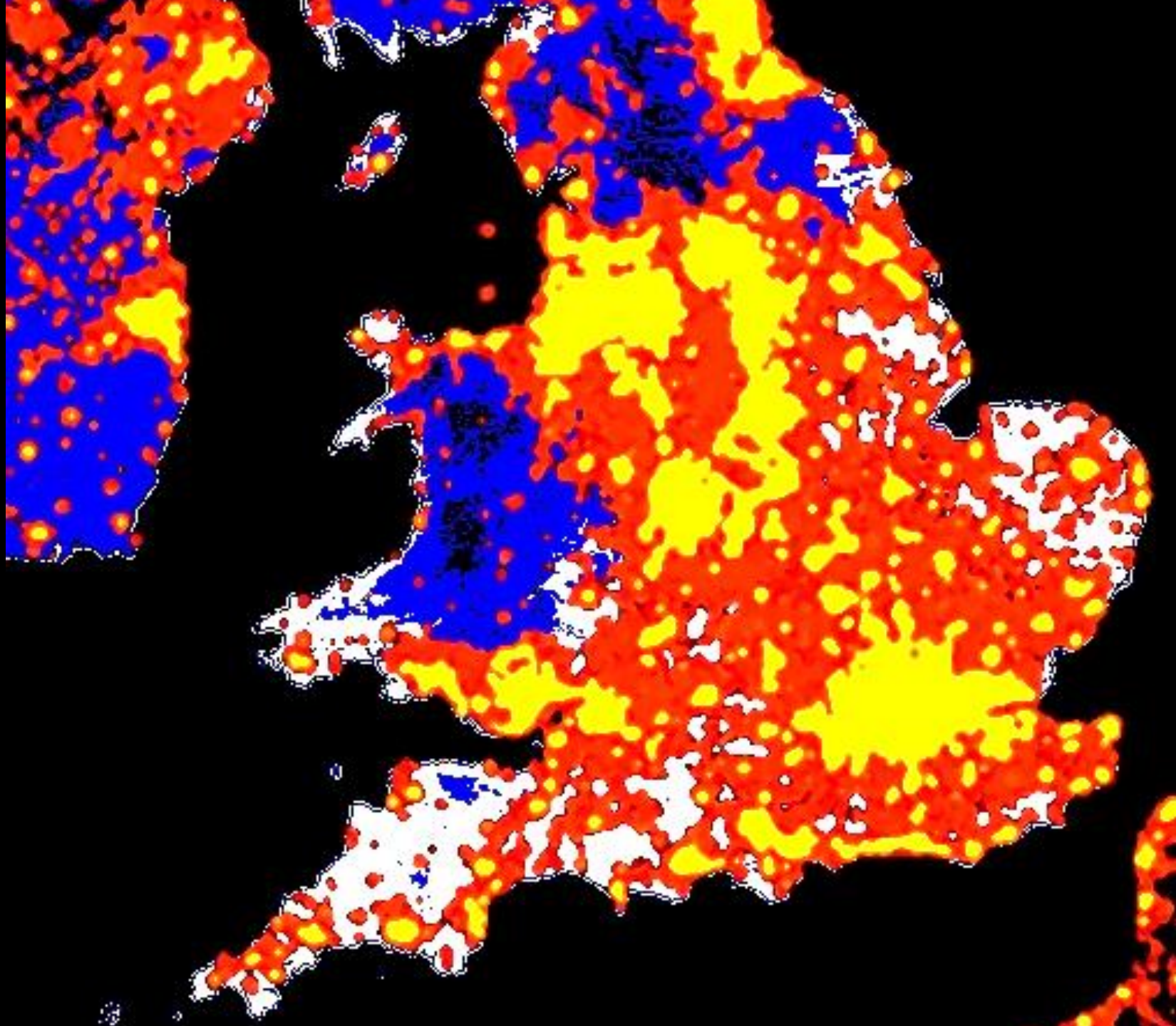
Light pollution



Light pollution



Light pollution



Method:

Camera settings

- f/number
- Exposure time
- ISO setting { [1/3] ISO is a standard measure of how a digital camera amplifies a signal (image) after a photograph is taken.

Method:

Camera settings

- f/number
- Exposure time
- ISO setting {
 - [2/3] Increasing the ISO amplifies the signal (and so brightness)...
 - ...but also amplifies the imperfections/noise too.

Method: Camera settings

- f/number
- Exposure time
- ISO setting { [3/3] These imperfections can be reduced by taking and removing a **dark frame**

ISO setting

ISO 100

ISO setting

ISO 200

ISO setting

ISO 400

ISO setting

ISO 800



ISO setting

ISO 1600

ISO setting

ISO 3200

Method: Camera settings

- f/number
- Exposure time
- ISO setting

The Moon



Exposure: Auto

ISO: Auto

Aperture: Auto



The Moon

Exposure: 3s

ISO: 3200

Aperture: f/7



Camera settings *for shooting the Moon*

- f/number { Anything from f/4.0 to f/32 !
- Exposure time { Usually short exposures
for the bright Moon:
1/50s-1/200 s
- ISO setting { The Moon is bright so ISO~200.
unless you want diffraction
spikes, then ISO1600+.

The Milky-way

A long-exposure photograph of a night sky. The Milky Way galaxy is visible as a bright, hazy band of light stretching diagonally across the frame from the bottom left towards the top right. In the foreground, a dark silhouette of a windmill stands on a grassy hill. To the left of the windmill, a row of small houses with red roofs is visible, their windows and some outdoor lights glowing with a warm orange light. The sky is filled with numerous stars, and the overall color palette is dominated by the dark blues and blacks of the night, the warm oranges of the artificial lights, and the pale, dusty colors of the Milky Way.

Best seen
in August
&
September



< Milky way

< Andromeda
galaxy



Milky way

< Andromeda
galaxy

Camera settings *for shooting the Milky Way*

- f/number { widest possible, e.g. f/4.0
- Exposure time { 30-60 seconds if
somewhere dark
- ISO setting { As high as your camera
can realistically do
e.g. ISO1600-4000+

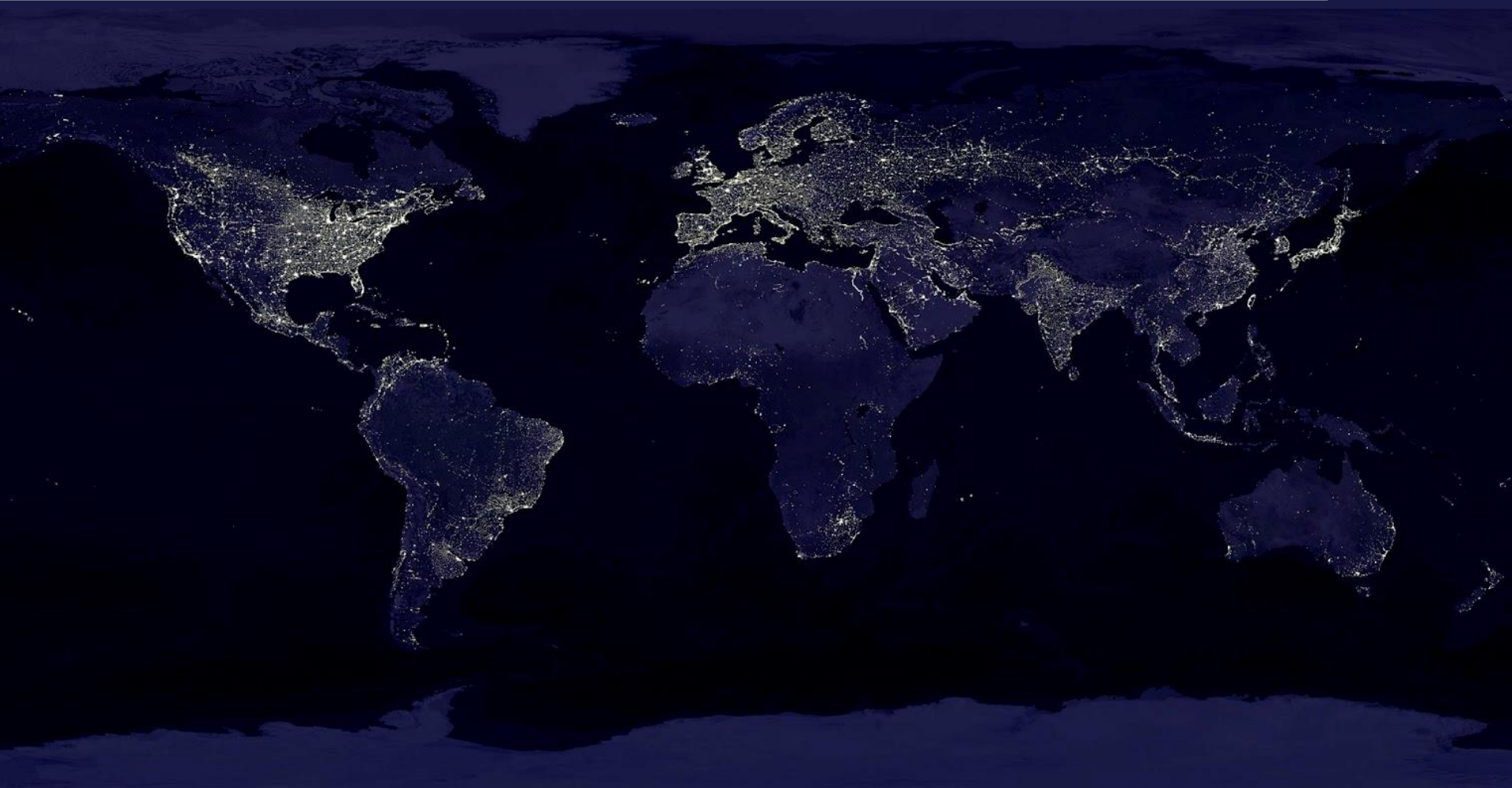
Camera settings

How to focus

Autofocus does not work at low light levels so...

- Use manual focus
- Focus on a bright star or planet
- ...or on a very distant light

International views



Travel for darker skies and different views



Utah, USA

Hawai'i

Brighton's horizon



Southern Hemisphere



Meteors





Meteors

Shooting stars are very difficult to photograph!

- They are very quick
- Often faint
- And can appear anywhere in the sky



Comet C/2025 A6 (Lemmon) October 2025

Canon 6D mk2 + 400mm lens
on a tracking mount.

42 x 10s exposures stacked
(7 minute exposure total).



Comets

Comets (and their dust)
are the source of
meteors (shooting stars)



A photograph of Comet F3 (NEOWISE) in a night sky. The comet is visible as a bright, elongated object with a distinct tail, positioned in the upper center of the frame. The sky is dark blue and filled with numerous stars. In the foreground, the silhouettes of trees and a church steeple are visible against the sky. The steeple is on the right side, and the trees are on the left and bottom. The text "Comet F3 (NEOWISE)" and "July 2020" is overlaid in the bottom left corner.

Comet F3 (NEOWISE)
July 2020

30 seconds
f/5.6
ISO 3200



Camera settings *for shooting stars*

- Exposure time { 5-15s until light pollution dominates
- ISO setting { Camera needs maximum sensitivity: ISO 3200+
- f/number { Wide open aperture
e.g. f/2.8 or f/4.0 to allow as much light in as possible

And take a large number of photos!

A dark night sky filled with numerous stars. A bright, thin streak of light, possibly a meteor or comet, extends diagonally from the upper right towards the center. The bottom of the image shows the dark silhouettes of trees and a fence line.

Next good
shower is
12th/13th
December

Spot the difference!



Aircraft/drones

- Have **flashing lights** to warn other aircraft.
- Take minutes to pass overhead.
- And they may change direction.



Satellites

- Usually **constant brightness**...
- ...although **sunlight may glint** (esp. for space debris) for a few seconds.
- Take minutes to pass.



Meteors

- Burn up in a second!
- **Uneven tracks as they explode** at the end of their journey.
- Shooting stars are often colourful.



Aurora

4 seconds
f/3.5
ISO 2500

Over Brighton
May 2024

Aurora



Over Brighton
May 2024

Aurora



Over Brighton
October 2024

Wide angle summary

- Settings
- Location
- Time of the year

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